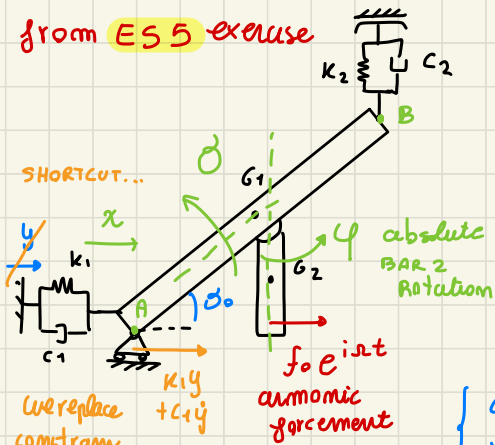


from ES5 exercise



⇒ We develop the matrix to solve on **MATLAB** we found $[\lambda_m]$...

$$\underline{x} = \begin{bmatrix} x \\ \phi \\ y \end{bmatrix}$$

replace y , because it is applied for a constraint where spring / damper are connected to system

{ as ground previously }

$$[\lambda_m] = \begin{bmatrix} 1 & -L_1 \sin(\theta_0) & 0 \\ 0 & L_1 \cos(\theta_0) & 0 \\ 0 & 1 & 1 \\ 1 & -L_1 \sin(\theta_0) & L_2 \\ 0 & L_1 \cos(\theta_0) & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{matrix} V_{G,x} \\ V_{G,y} \\ u_1 \\ V_{G_2,x} \\ V_{G_2,y} \\ u_2 \end{matrix}$$

uncoupled relationship between δl and indep coord

$$[\lambda_k] = \begin{bmatrix} +1 & 0 & 0 \\ 0 & 1 & -2L_1 \cos(\theta_0) \\ 0 & 0 & 1 \end{bmatrix} \begin{matrix} \delta l_1 \\ \delta l_2 \end{matrix}$$

here we have now linear kin relationship between δl and u and also for h so we have also K_{II}, K_g

$$K_{el II,2} = K_2 \delta l_{02} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 2L_1 \sin(\theta_0) & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$[K_g] = m_1 g \begin{bmatrix} 0 & 0 & 0 \\ 0 & -L_1 \sin(\theta_0) & 0 \\ 0 & 0 & 0 \end{bmatrix} + m_2 g \begin{bmatrix} 0 & 0 & 0 \\ 0 & -L_1 \sin(\theta_0) & 0 \\ 0 & 0 & L_2 \end{bmatrix}$$

$[K_{g1}]$
BODY 1
 $[K_{g2}]$
BODY 2

no effect now associated to coord x !

$[m], [c], [k]$

all obtained by hand → we can use this matrix to obtain ...

We obviously need numerical DATA to solve the problem:

$$m_1 = 10 \text{ kg} \quad J_1 = 0,8 \text{ kg m}^2 \quad L_1 = 0,5 \text{ m}$$

$$\theta_0 = \pi/6$$

$$m_2 = 3 \text{ kg} \quad J_2 = 0,2 \text{ kg m}^2 \quad L_2 = 0,2 \text{ m}$$

$$K_1 = 500 \text{ N/m}$$

$$C_1 = 0,2 \text{ N}\cdot\text{s/m}$$

$$K_2 = 500 \text{ N/m}$$

$$\Delta l_{0,2} = 0,1275 \text{ m}$$

$$C_2 = 0,2 \text{ N}\cdot\text{s/m}$$

affecting 2nd ORD el stiff



working in

MATLAB!

using ground MATRIX

$$[m] = [\lambda m]^T [m_{ph}] [\lambda m]$$

$$[K] = \underbrace{[\lambda k]^T [K_{ph}] [\lambda k]}_{[K_I]} + \underbrace{[K_{el II 2}]}_{\substack{2^{nd} \text{ ORD} \\ EL}} + \underbrace{[K_{g_1}] + [K_{g_2}]}_{[K_g]} \quad \begin{matrix} \text{Body 1} & \text{Body 2} \end{matrix}$$

$$[C] = [\lambda k]^T [C_{ph}] [\lambda k] \quad (\text{keeping } [\lambda k] = [\lambda c]) \quad C // K!$$

>> "1" for the TRANSPOSE

- peculiarity: $[K]$ is Diagonal matrix!

There are independent effect on elongation...

NO coupling through $[K]$ terms → all obviously POSITIVE
(positive definite $[K]$)

- again also $[C]$ is Diagonal and because φ does not contribute $[\lambda k] \rightarrow$ no damper effect due to $\varphi \Rightarrow C(3,3) = 0$
so pendulum
Rotation NOT damped!

$\omega, \underline{x}$? $\rightarrow$ we know this correspond to solve

$$(\underbrace{\omega^2}_{\text{modal frequencies}}) \mathbf{I} - [\mathbf{m}]^{-1} [\mathbf{K}] \underbrace{(\underline{x})}_{\substack{\text{modal} \\ \text{shape vector } \forall \omega_i}} = 0$$

Eig value / vector problem $\uparrow$ compare this ...

$$(\underbrace{\lambda \mathbf{I} - \mathbf{A}}_{\substack{\text{eig} \\ \text{values of } \mathbf{A}}}) \underbrace{(\underline{x})}_{\substack{\text{eig vectors} \\ \text{of } \mathbf{A} \text{ corresp.}}} = 0$$

$$\text{if } [\mathbf{A}] = [\mathbf{m}]^{-1} [\mathbf{K}]$$

$\downarrow$
extracting eig value / vectors
 $\downarrow \quad \downarrow$
 $\omega^2 \quad \underline{x}$

(normalized)
as $\sum x_i^2 = 1$

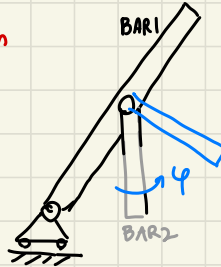
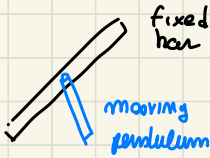
$\rightarrow$ on MATLAB!

obtaining modal shape for ω_1

$$\underline{V}_1 = \begin{Bmatrix} 0.0396 \\ -0.0155 \\ \underline{0.9991} \end{Bmatrix} \begin{matrix} x \\ \phi \\ y \end{matrix}$$

In mode 1 of motion @ ω_1 :
@ $f_1 = 0.6563 \text{ Hz}$

$\uparrow$
higher relevant term



the BAR1 stay still almost
will vertical BAR2
(oscillating (pendulum))

mean to pendulum motion
fig (similar to a simple
pendulum)

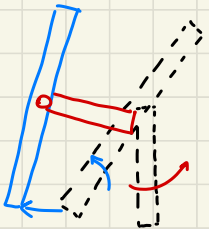
$\uparrow$ natural freq
of an hung = d bar

$$\underline{V}_2 = \begin{Bmatrix} -0.2479 \\ 0.1271 \\ \underline{0.9604} \end{Bmatrix} \begin{matrix} x \\ \phi \\ y \end{matrix}$$

@ $f_2 = 1.913 \text{ Hz}$

equilibrium pos

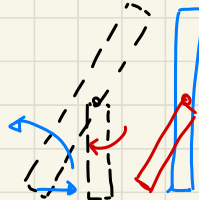
visualization
of $\underline{V}_2$ happening



for undamped
system

While $\underline{V}_3 = \begin{pmatrix} 0.3385 \\ \underline{0.9091} \\ -0.2428 \end{pmatrix}$ $\begin{matrix} \text{large rotation} \\ \text{neg rotation} \end{matrix}$

@ $f_3 = 1.81 \text{ Hz}$



- considering also the forcing effects (come @ time) → using super-imposition we can find the overall

$$\text{from } [\lambda_f] = \begin{bmatrix} 1 & -L_1 \sin(\theta_0) & 2L_2 \\ 1 & 0 & 0 \end{bmatrix}$$

$$\underline{f}_{ph} = \begin{pmatrix} f_0 e^{i\omega t} \\ k_1 y + c_1 \dot{y} \end{pmatrix}$$

the **FRF** is the solution of the problem $(-\omega^2[m] + i\omega[c] + [k]) \underline{x}_0 = \underline{F}_0$ excitation

$$(-\omega^2[m] + i\omega[c] + [k])^{-1} = G(i\omega)$$

found by solving the (*)

problem with $\underline{F}_0 = [k_k]^T [i]$ equally for $\underline{f}_{ph} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

$$\underline{f}_{ph} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

apply unit input along horiz force

$$[\lambda_f]^T \underline{f}_{ph}$$

superimposition approach...

we select from

from the found **DIAGRAM BODE**

you can see it is low damp because the pick is very high and the phase transition is very rapid from 0 to π

and 3rd resonance has no so much excitation @ $f = 1.81 \text{ Hz}$ small resonance

probably because not so much motion of lower tip pendulum when $\theta > 0$ IT IS LIKE A NODAL POINT

↓
no resonance pick @ f_3 , while big resonance @ f_1, f_2

for the second force we consider an input on point A

$$ky + c\dot{y} = \bar{F}$$

↓

as input we have the displacement y (NOT a force)
so we should set $y = 1$ (amplitude of motion undamped)

so of setting

$$y = y_0 e^{i\Omega t} \text{ and } y_0 = 1 \text{ unitary}$$

$$\dot{y} = i\Omega y_0 e^{i\Omega t} \rightarrow \text{equivalent force } F =$$
$$ky + c\dot{y} = y_0 e^{i\Omega t} (K + i\Omega C)$$

depends on freq.
exp. function
magnitude of force $y_0 = 1$
complex value

so the force is not maximum
magnitude when y is max (because of Im part)

and magnitude depends on $\Omega \rightarrow$ so it must be update inside
for cycle on matlab

respect the constrain displacement

excitement Bode plot, we get 3 different peaks @ f_1, f_2, f_3

↓

we can interpret physically this Bode diag

how to handle this

- QUESTION 3: find the FRF respect output of a constrain force ...
to extract from the syst.